

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA02

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO3: Solve problems for searching, sorting, and merging using array and string processing techniques. (BL3, BL4)

Assessment Tool Weightage: 40%

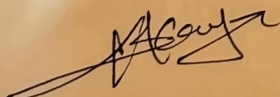
QUESTIONS

Total Marks:100

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I K.Kavya-19242523 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

90%
✓

Maximum product subarray:

aim

to develop a program that finds the contiguous subarray with the maximum product while handling positive and negative numbers efficiently.

problem understanding (20%)

the system must:

- identify the contiguous subarray with the highest product
- * handle negative values carefully because multiplying two negative numbers gives a positive result
- * track both maximum & minimum products dynamically

method selection (20%)

Dynamic programming approach is used

→ maintain:

maxprod → maximum product ending at current index

minprod → minimum product ending at current index

- * this method efficiently handles negative values using dynamic tracking

Interpretation of results (15%)

- * the program successfully identifies the contiguous subarray with the maximum product
- * Efficiently handles negative values using dynamic tracking.

87. Remove all occurrence of character

Aim:

to create a program that removes all occurrence of a specified character from a string efficiently.

problem understanding (20%)

the system must:

- * Remove every occurrence of a given character
- * Preserve the order of remaining character
- * Perform operation efficiently in-place.

method selection (20%)

- * two-pointer technique is used
- * one pointer reads character, another writes valid character

Interpretation of results (15%)

- * the program removes all specified character correctly
- * original order of remaining character is preserved

complexity: $O(n)$

8. Union of two arrays

Aim:

to combine two arrays and produce a collection containing any unique elements

problem understanding

the system must:

- * merge two datasets
- * Remove duplicate elements

preserve all unique data values
method selection

- * Nested loop duplicate checking method is used.
 - * simple and suitable for moderate input series
- interpretation of results

- * the program successfully combines both arrays
- * Duplicate values are removed
- * ensure all unique elements are retained

89 count inversion in an array

Aim:

to determine how far an array is from being sorted by

counting inversion pair

problem understanding

- * the system must count pairs (i, j) where $i < j$ and $arr[i] > arr[j]$
- * efficiently compare ordering using optimized technique

method selection

- * modified merge sort is used
- * During merging, inversion are counted efficiently

interpretation of result

- * the inversion count indicates how unsorted the array is higher inversion count means greater disorder time complexity : $O(n \log n)$

First Repeating element:

aim:

to identify the first repeating element in an array while preserving element order.

problem understanding:

the system must:

- * Detect repeated elements
- * Return the first repeating element based on occurrence order
- * efficiently track visited elements

method selection:

- * Nested loop comparison method is used
- * check occurrence order carefully

interpretation of results:

- * the program correctly identifies the earliest repeating element
- * Preserve original order while checking repeating
- * useful in log analysis and duplicate detection systems

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation